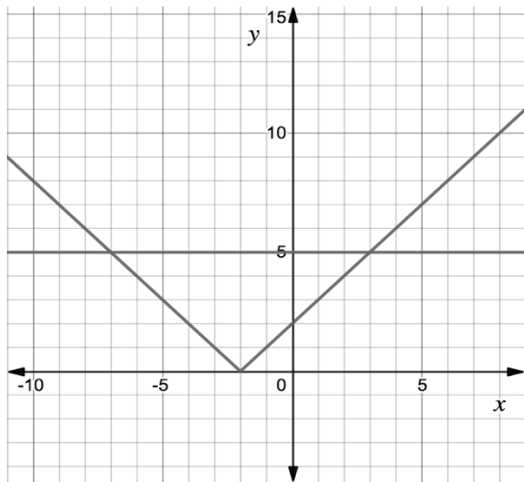


Algebra 1
Unit 10
Lesson 3 Practice

Name _____
Date _____
Period _____

For questions 1-2, consider the absolute value equation $|x + 2| = 5$. The graphs of $y = |x + 2|$ and $y = 5$ are shown on the coordinate grid.



1. Use the graph to determine the solution to $|x + 2| = 5$.

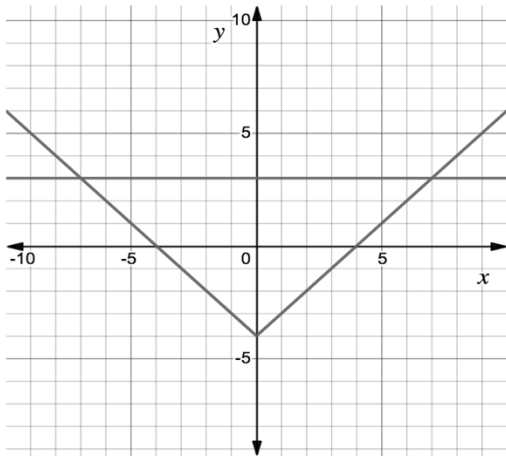
$x = 3$ and $x = -7$

2. Solve $|x + 2| = 5$ algebraically.

$x + 2 = 5$
 $x = 3$

$x + 2 = -5$
 $x = -7$

For questions 3-4, consider the absolute value equation $|x| - 4 = 3$. The graphs of $y = |x| - 4$ and $y = 3$ are shown on the coordinate grid.



3. Use the graph to determine the solution to $|x| - 4 = 3$.

$$x = 7 \text{ and } x = -7$$

4. Solve $|x| - 4 = 3$ algebraically.

$$\begin{array}{ll} |x| - 4 = 3 & |x| = 7 \\ x = 7 & x = -7 \end{array}$$

For questions 5-10, solve each absolute value equation.

5. $|m| - 37 = -12$

$$\begin{array}{l} +37 \quad +37 \\ |m| = 25 \\ m = 25 \quad m = -25 \end{array}$$

6. $|h - 19| = 34$

$$\begin{array}{ll} h - 19 = 34 & h - 19 = -34 \\ h = 53 & h = -15 \end{array}$$

7. $|4x| = 22$

$$\begin{array}{ll} 4x = 22 & 4x = -22 \\ x = 5.5 & x = -5.5 \end{array}$$

8. $21 = |x| + 9$

$$\begin{array}{l} 12 = |x| \\ x = 12 \quad x = -12 \end{array}$$

9. $|k - 49| = 68$

$$\begin{array}{ll} k - 49 = 68 & k - 49 = -68 \\ k = 117 & k = -19 \end{array}$$

10. $21 = |9x|$

$$\begin{array}{ll} 9x = 21 & 9x = -21 \\ x = 2.\bar{3} & x = -2.\bar{3} \end{array}$$